Materials and light

The word search contains 18 words used when describing the effect of light on materials. Look for the words listed below.

s	e	r	d	s	m	o	o	t	h	t	d	r	y	m
f	j	o	d	a	o	r	s	t	g	r	l	u	n	o
t	n	u	n	u	r	g	c	r	m	a	e	j	e	s
m	h	g	e	v	s	k	a	a	a	n	f	s	f	h
n	w	h	r	u	b	t	t	n	t	s	q	u	a	i
d	s	e	n	s	o	r	t	s	e	p	i	r	n	n
a	q	q	l	d	q	f	e	m	r	a	r	f	c	y
b	r	q	a	i	b	x	r	i	i	r	j	a	j	g
s	e	r	o	s	g	g	t	s	a	e	d	c	x	l
o	f	m	p	i	u	h	m	s	l	n	s	e	f	a
r	l	y	a	n	b	v	t	i	r	t	g	t	j	s
b	e	n	q	t	t	i	e	o	r	u	i	x	l	s
w	c	o	u	a	t	a	g	n	t	r	v	b	v	o
e	t	v	e	e	i	r	h	k	c	a	o	b	l	s
e	t	r	a	n	s	l	u	c	e	n	t	r	s	h

Here are the words:

absorb
dark
glass
light
material
matt
mirror
opaque
reflect
rough
scatter
sensor
shiny
smooth
surface
transmission
translucent
transparent

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Can light be bent?

Solve the clues and write the answers in the answer grid below. If you get all the answers correct the first vertical column will show you another word connected with this topic.

Clues

1 A natural light source.

2 A transparent material, similar to glass.

3 Something you do in the science lab to find out how things behave.

4 '**R**ichard **o**f **Y**ork **g**ained **b**attle **i**n **v**ain.' This rhyme helps you to remember the _____ in a rainbow.

5 The shape of a glass prism used to show dispersion of light.

6 This happens to light as it passes from air into water.

7 The direction in which light travels to your eye when you look at the bottom of a glass.

8 Light refracts when it passes from one _____ to another.

Answer grid

1										
2										
3										
4										
5										
6										
7										
8										

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How can we change colour?

Complete these sentences using the colours in the box to fill in the gaps. You may use each colour once, more than once or not at all.

black	blue	cyan	green	magenta	red	white	yellow

1 If we shine white light through a red filter, only _____ light comes through. The red filter transmits only _____ light and absorbs all the other colours.

2 The primary colours are _____, _____ and _____.

3 When the three primary colours are mixed together we get _____ light.

4 Red and green filters are used together to get _____ light.

5 _____ and _____ filters must be used together to get cyan.

6 A blue ball looks blue because it is reflecting mostly _____ light.

7 If we look at the blue ball in blue light, it looks _____.

8 If we look at the blue ball in green light, it looks _____.

9 If we look at a yellow ball in green light, it looks _____.

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